Jiaqi Kevin Yang

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EDUCATION

MSc in Computer Science at McGill University **Sep 2026 - Dec 2027**

- Teaching Assistant for **200+** students in assembly programming and computer architecture

BSc in Software Engineering at McGill University **Sep 2022 - May 2026**

- Cumulative GPA: 3.74/4.00

EXPERIENCE

QA Automation Intern at dormakaba **May 2025 - Aug 2025**

- Built a custom Locust load testing framework that simulated up to **1000** concurrent users and tested how the backend API behaved under different stress patterns to uncover bottlenecks and scalability issues
- Improved integration tests via a Flask-based internal API, reducing manual QA by **~20** hours per release cycle using a physical testing jig powered by a Raspberry Pi module to validate smart lock functions
- Integrated over **50** Java-based end-to-end validation tests into CI/CD pipelines to catch bugs before release
- Implemented dynamic secret management across **3** repositories using AWS SSO and Parameter Store to prevent leaking API keys and user secrets

PROJECTS

C Compiler **Jan 2026 - May 2026**

- Built a compiler in Java that translated the C programming language into MIPS assembly
- Refactored C grammar to eliminate left recursion and built a recursive-descent parser to generate syntax trees
- Engineered multi-pass semantic analysis engine to enforce strong typing, nested scoping and struct alignment
- Generated target-specific function calling conventions, stack frame layout management, and jump offsets

Smart Dumbbell **Nov 2025**

- Engineered a non-blocking, multithreaded Flask backend to process concurrent MQTT hardware streams and a computer vision pipeline to accurately detect and classify different workout exercises
- Served as a technical lead for backend and systems integration, coordinating hardware and software teams
- Designed a JavaScript/HTML frontend to display live statistics, end-of-set encouragement and form corrections

Ataxx Game Playing AI **Nov 2025 - Dec 2025**

- Engineered an adversarial game AI using minimax search with alpha-beta pruning and iterative deepening
- Implemented heuristic move sorting, cutting depth-4 search latency by nearly **50%** to allow deeper searches
- Reduced search-tree branching by **33%** by applying move folding logic that pruned redundant state transitions

AI Study Assistant **Nov 2024**

- Deployed a local Qwen 2.5 pipeline, integrating a large language model into a full-stack Django web app to generate various study tools such as summaries, flashcards, study podcasts and quizzes
- Integrated OpenAI Whisper, enabling real-time audio transcriptions for the model to process lecture recordings

SKILLS

Programming Languages: Java, Python, C#, JavaScript/TypeScript, HTML, C, C++, Bash, SQL, GraphQL
Technical Skills: Git, Agile, Linux/Unix, Docker, Cursor, REST APIs, AWS, Kubernetes, PyTorch
CS Fundamentals: Operating Systems, Distributed Systems, Concurrent Programming, Machine Learning
Languages: Fluent English, Fluent French, Advanced Mandarin Chinese, Beginner Japanese